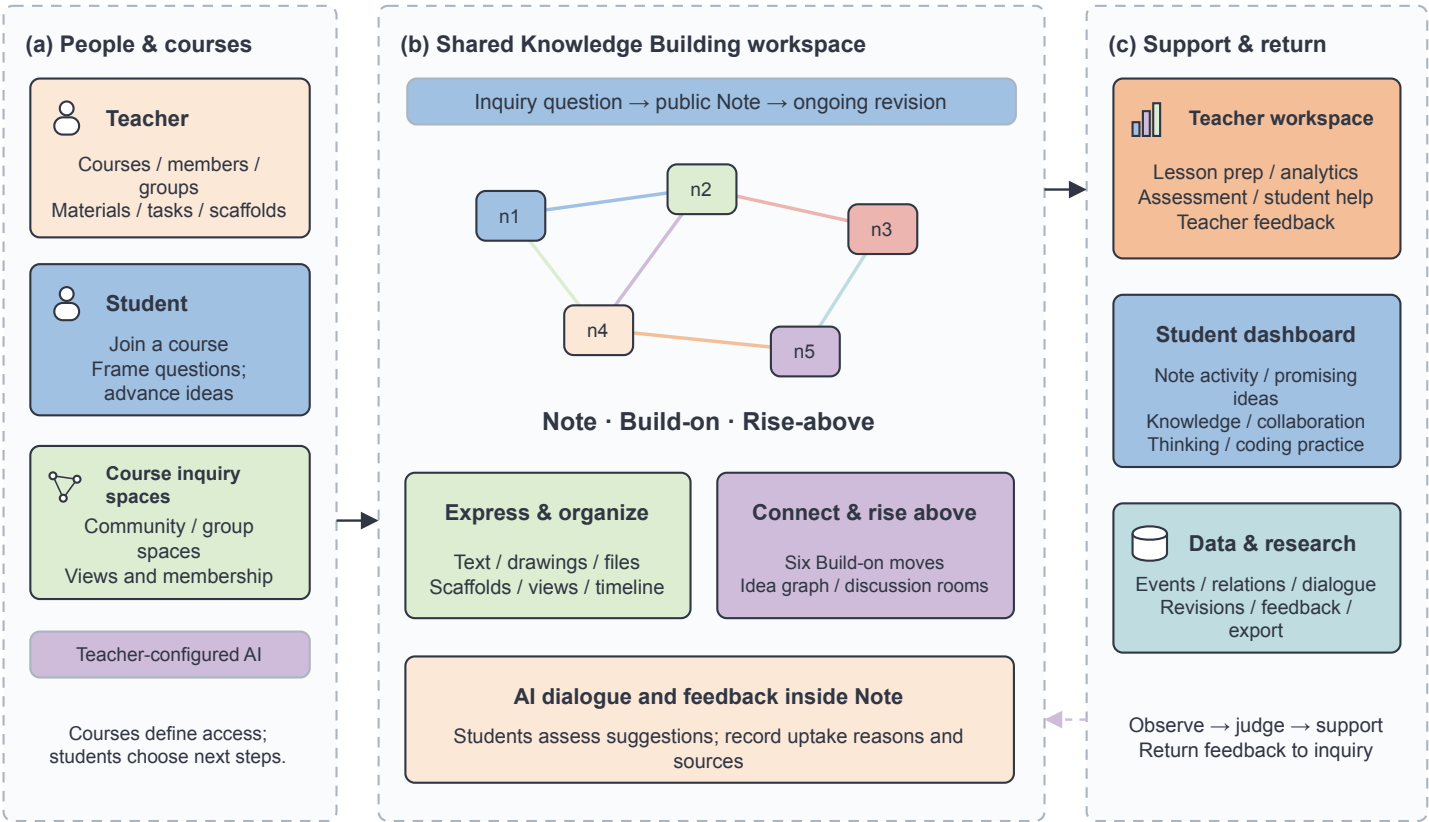


HAKCC: system functions and collaboration

Public ideas connect student inquiry, teacher support, and AI collaboration



Colors distinguish functional layers; arrows show information or collaboration, not mandatory steps.

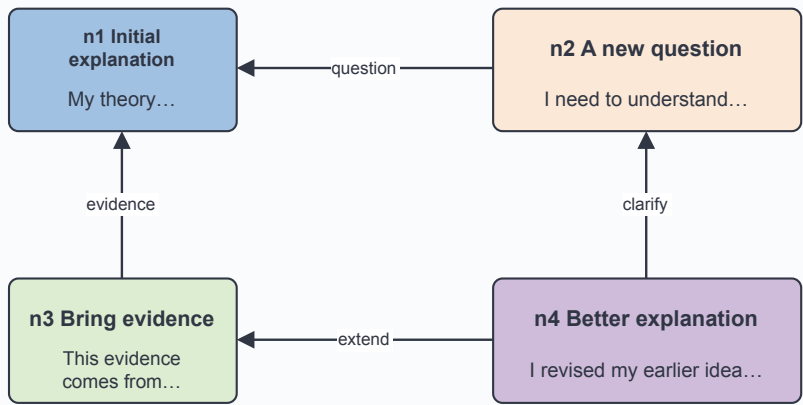
How public ideas develop over time

Shared questions, Build-on moves, evidence, and revisions advance community knowledge

(a) Questions and multimodal contributions

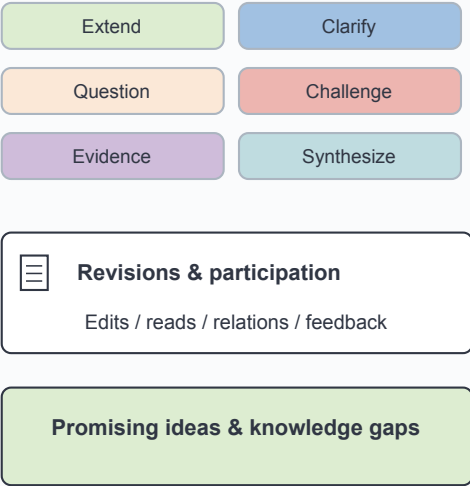


(b) Public, improvable idea network



Each new Note retains its author and move;
public ideas invite peer response, revision, and reinterpretation.

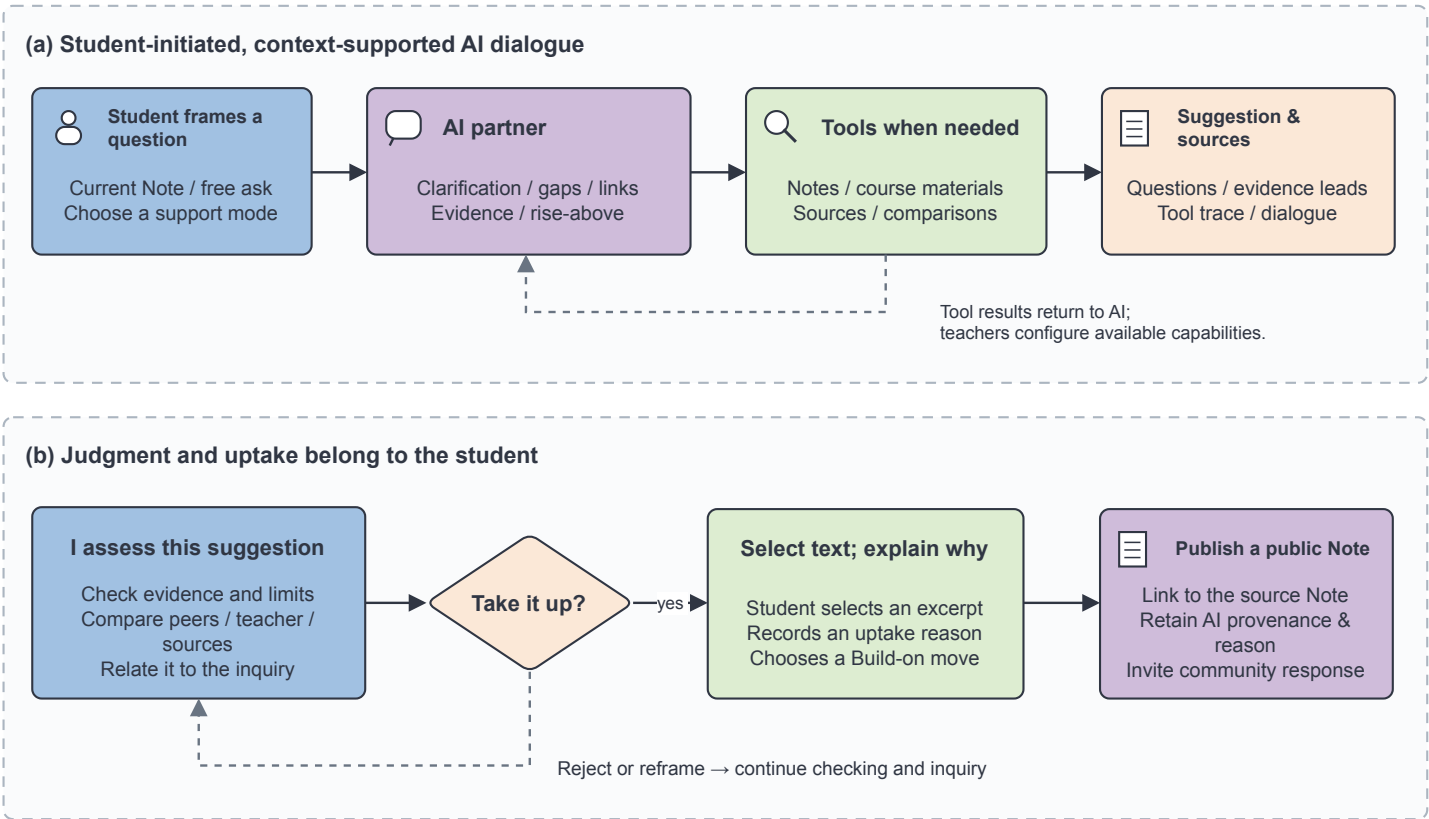
(c) Moves and traceability



Illustrated direction matches storage: new Note → prior Note being addressed. Colors are figure-specific.

AI partnership and student uptake

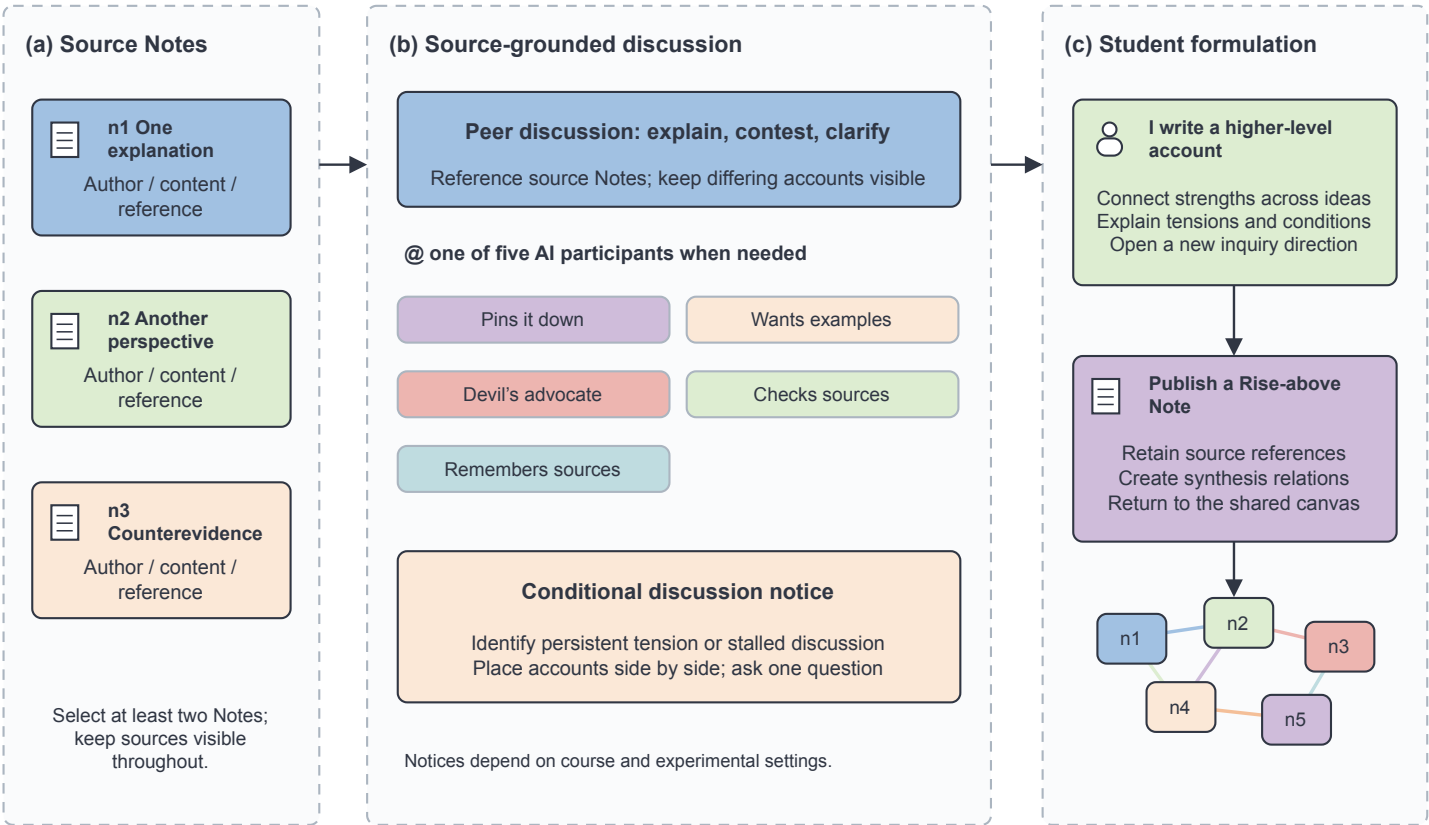
AI offers checkable suggestions; students retain judgment, choice, and public authorship



Publishing an AI excerpt requires a reason; the diagram does not imply that every AI reply must be adopted.

From diverse ideas to student-authored Rise-above

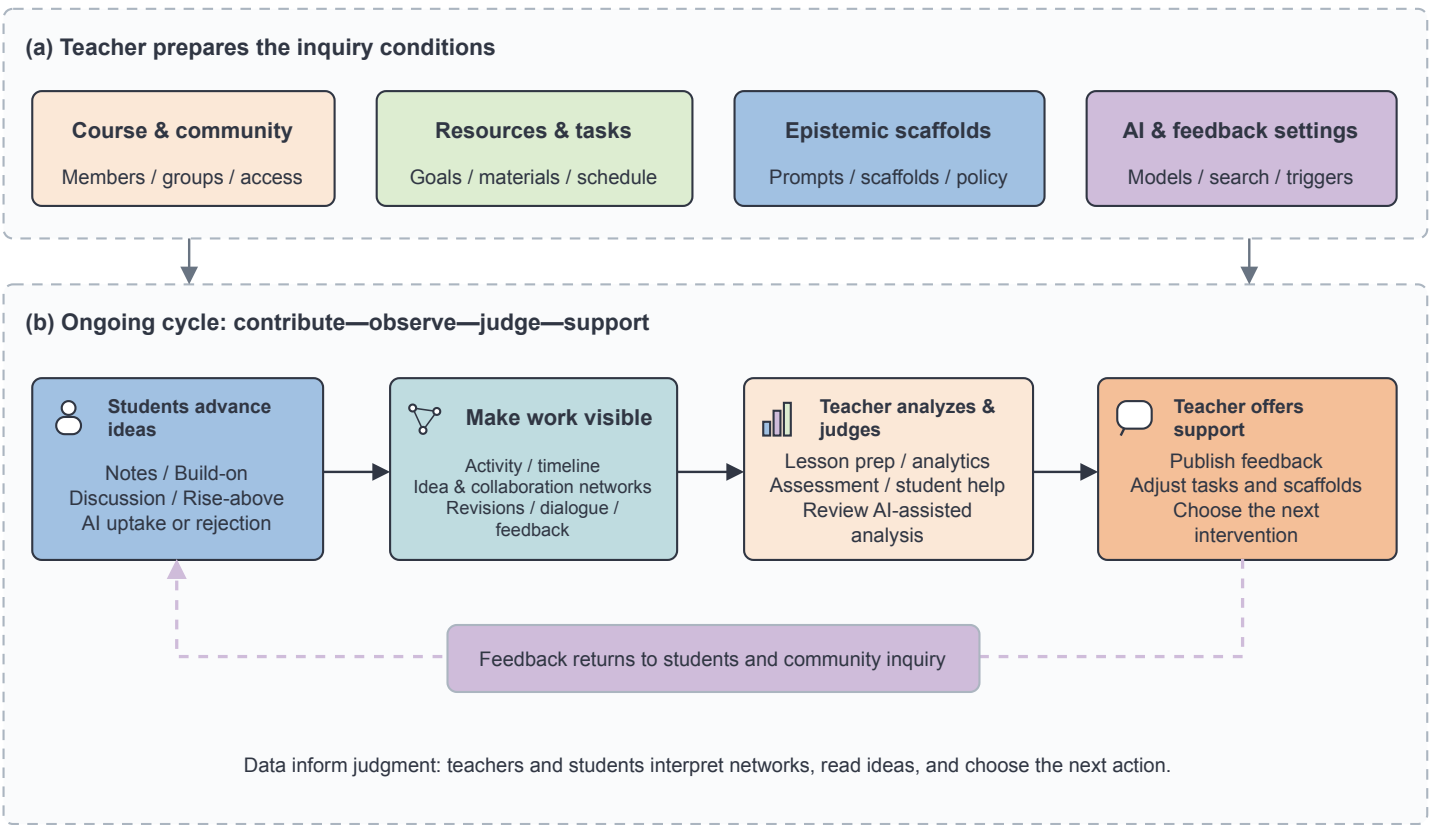
Persistent sources, peer negotiation, optional AI participation; students write the higher-level account



AI participants contribute questions, examples, counterexamples, sources, and links; published text is student input.

Teacher support and the student inquiry loop

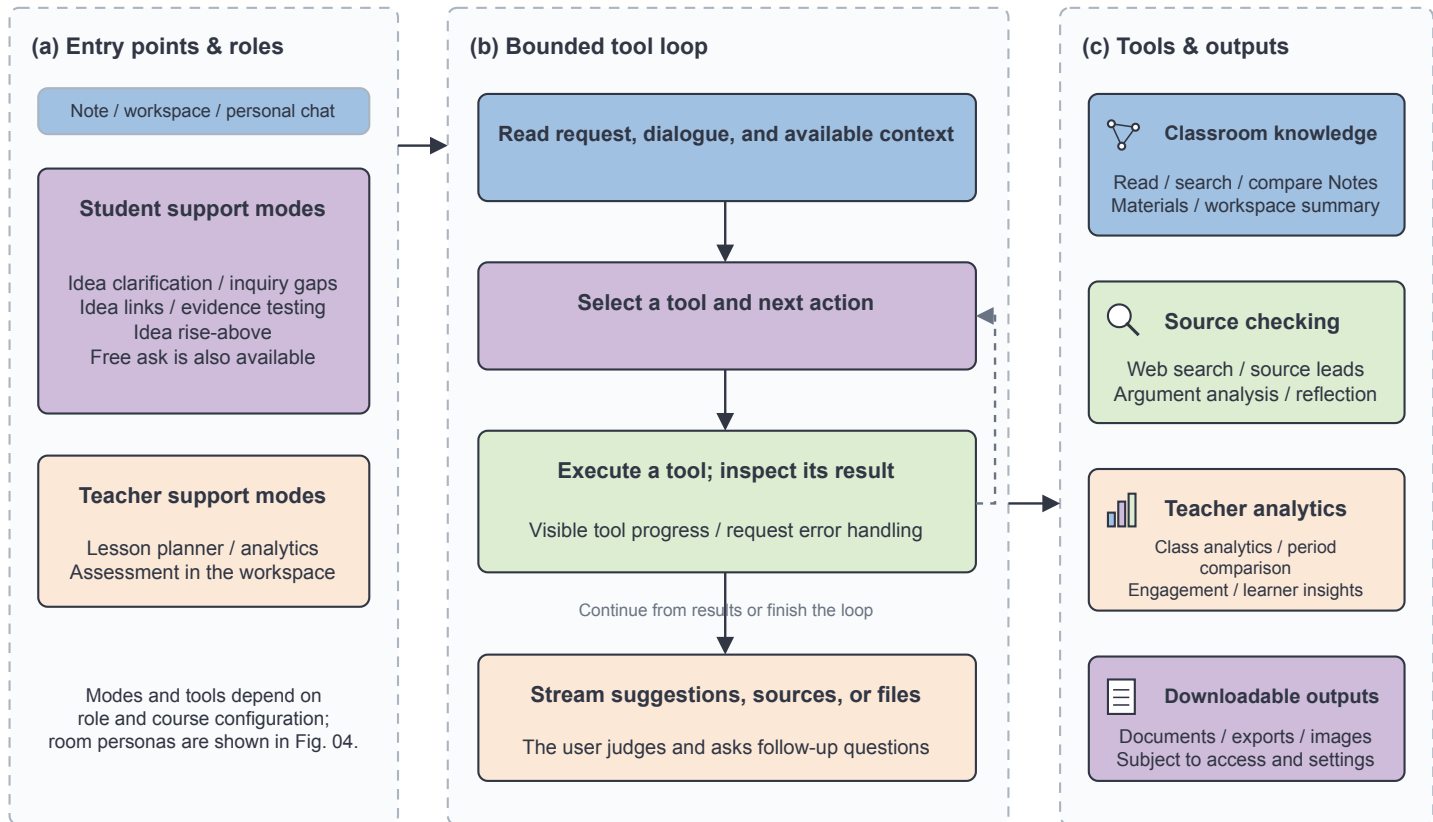
Observe community knowledge work, judge when to intervene, return feedback to idea improvement



Activity, networks, dialogue, and revisions inform teacher judgment; feedback supports the next inquiry action.

Pedagogical AI roles and tool use

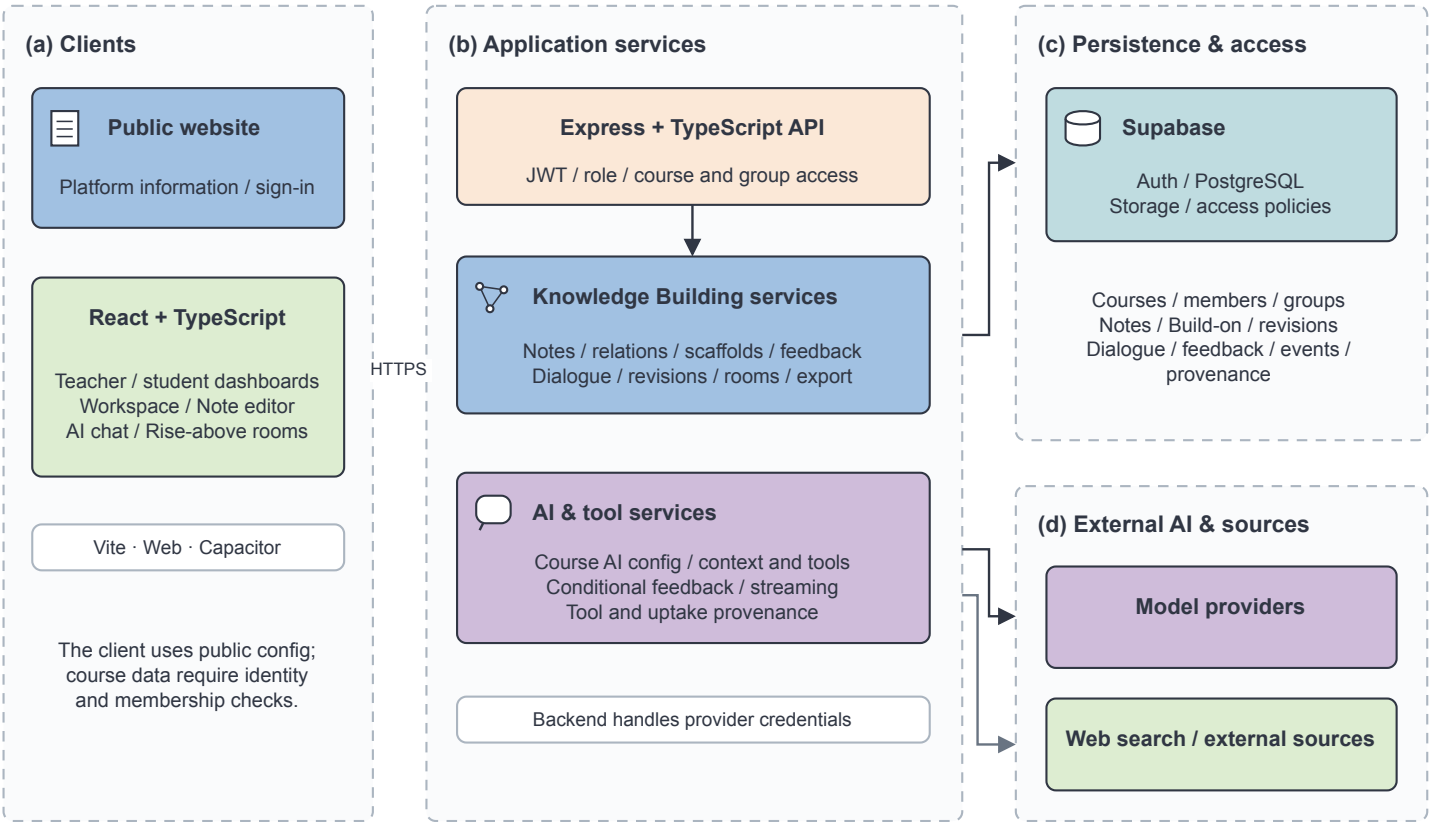
Requests and roles select tools within accessible course context



The diagram shows a bounded loop within a request; tools vary by role, course settings, and entry point.

HAKCC architecture and data boundaries

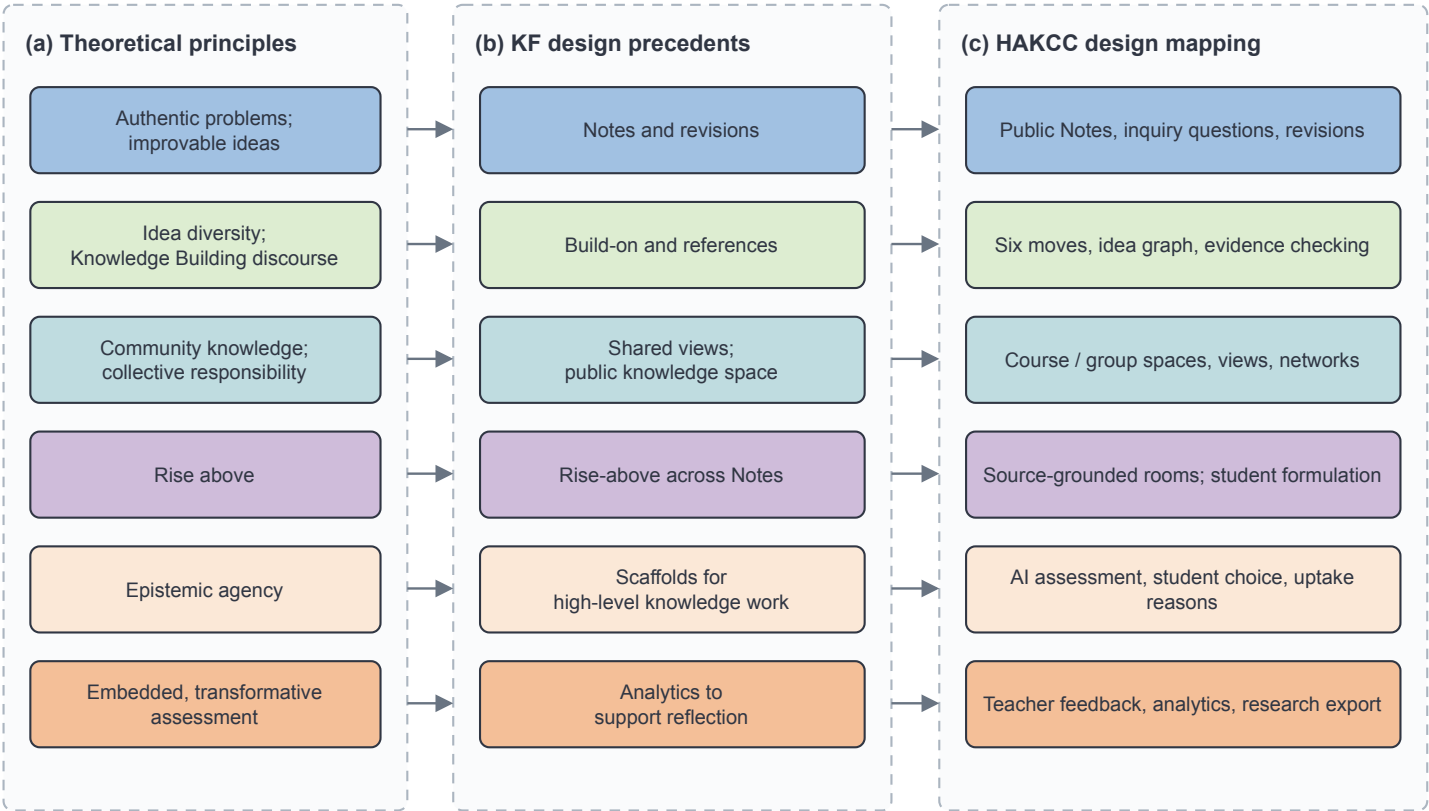
Public interface, course access, backend AI calls, and traceable knowledge data



Solid lines show request or data paths; AI and search use the backend; course data require membership checks.

From Knowledge Building theory to HAKCC design

Informed by Scardamalia and Bereiter's theory and Knowledge Forum's design

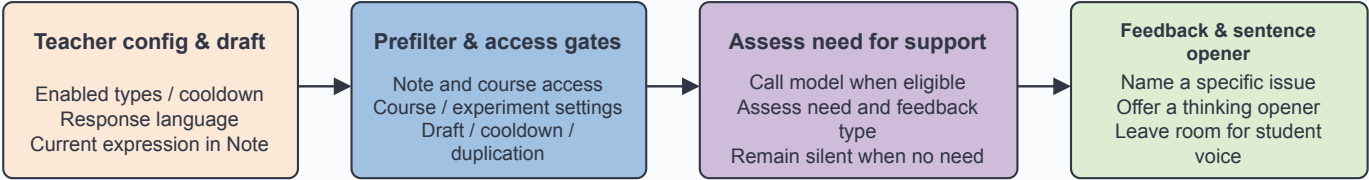


Mapping is interpretive, not outcome validation. Theory and KF precedents: Tarchi et al. (2013); see figure notes.

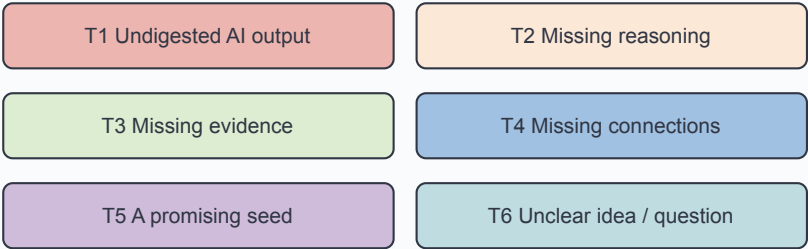
Conditional AI feedback and adaptive scaffolds

Detect where the current idea needs development; short feedback supports the student's next move

(a) Course settings and the current Note govern feedback

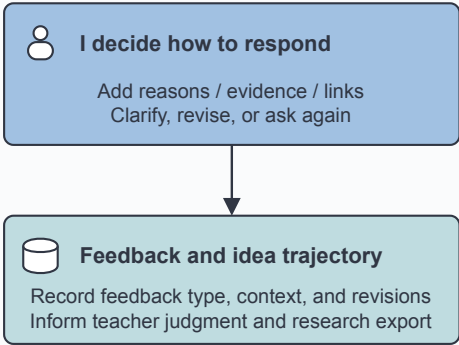


(b) Six feedback concerns: T1–T6



Feedback identifies an opportunity to advance; T5 develops an already promising idea.

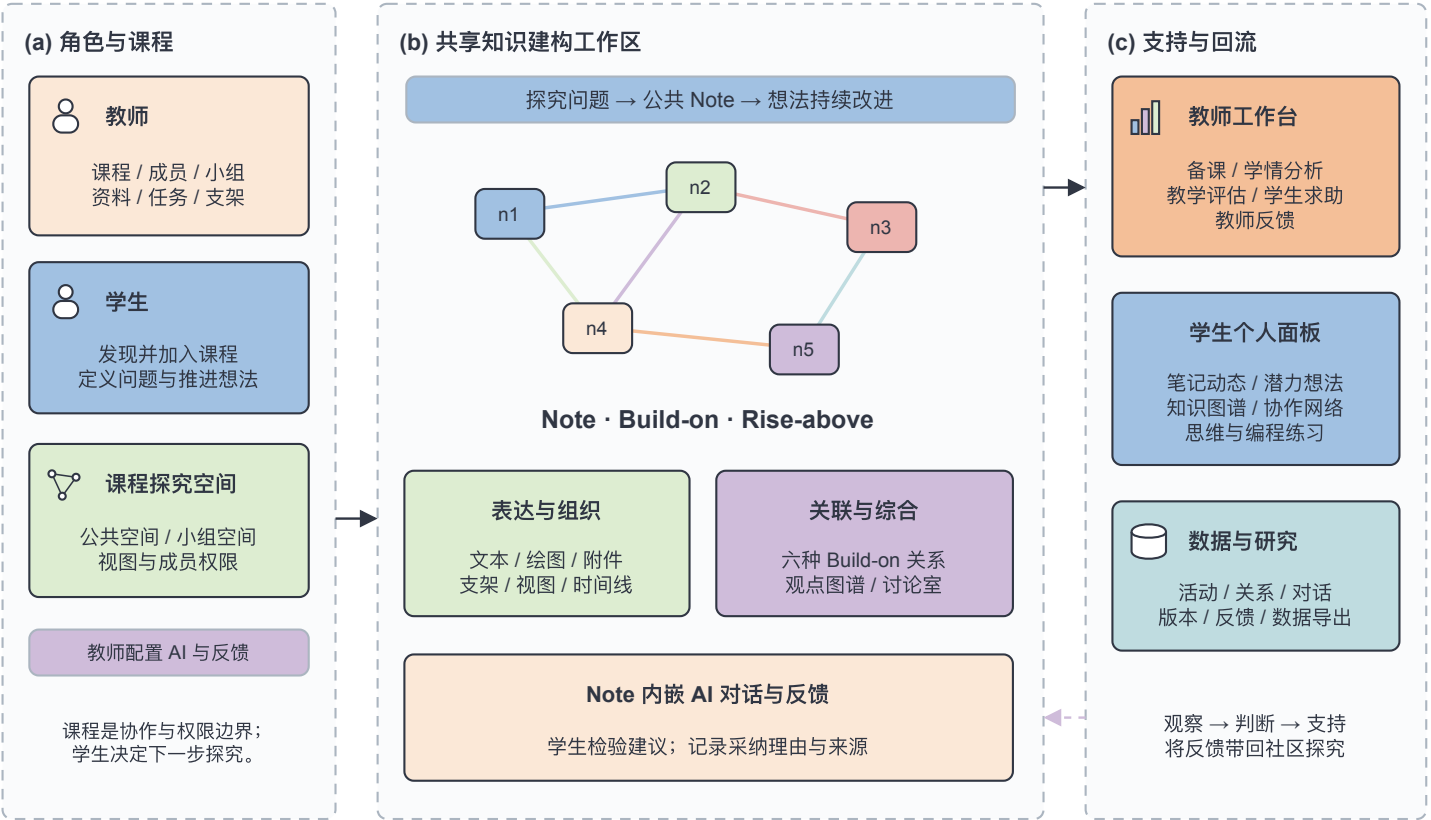
(c) Student response & traceability



Automatic checks and requested feedback use different entry points; availability depends on settings and the request.

HAKCC：系统功能与协作机制

以公共想法为中心，连接学生探究、教师支持与 AI 协作



颜色区分功能层；箭头表示信息或协作流，非强制学习步骤。

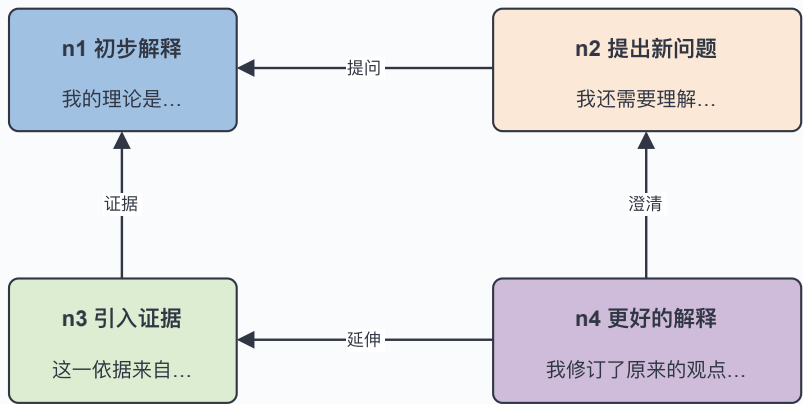
公共想法如何持续改进

从共同问题出发，通过 Build-on、证据与修订推进社区知识

(a) 问题与多模态贡献

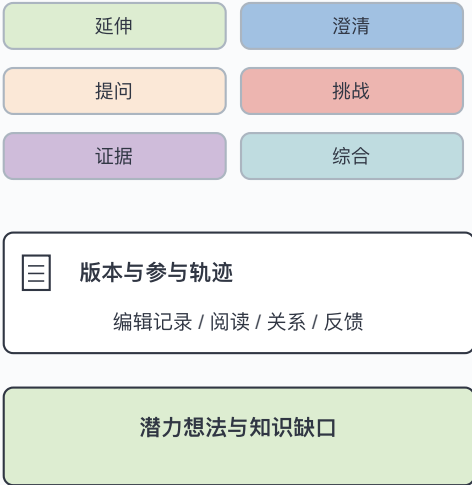


(b) 可改进的公共想法网络



每条新 Note 保留作者与关系意图；
公共想法接受同伴回应、修订和再解释。

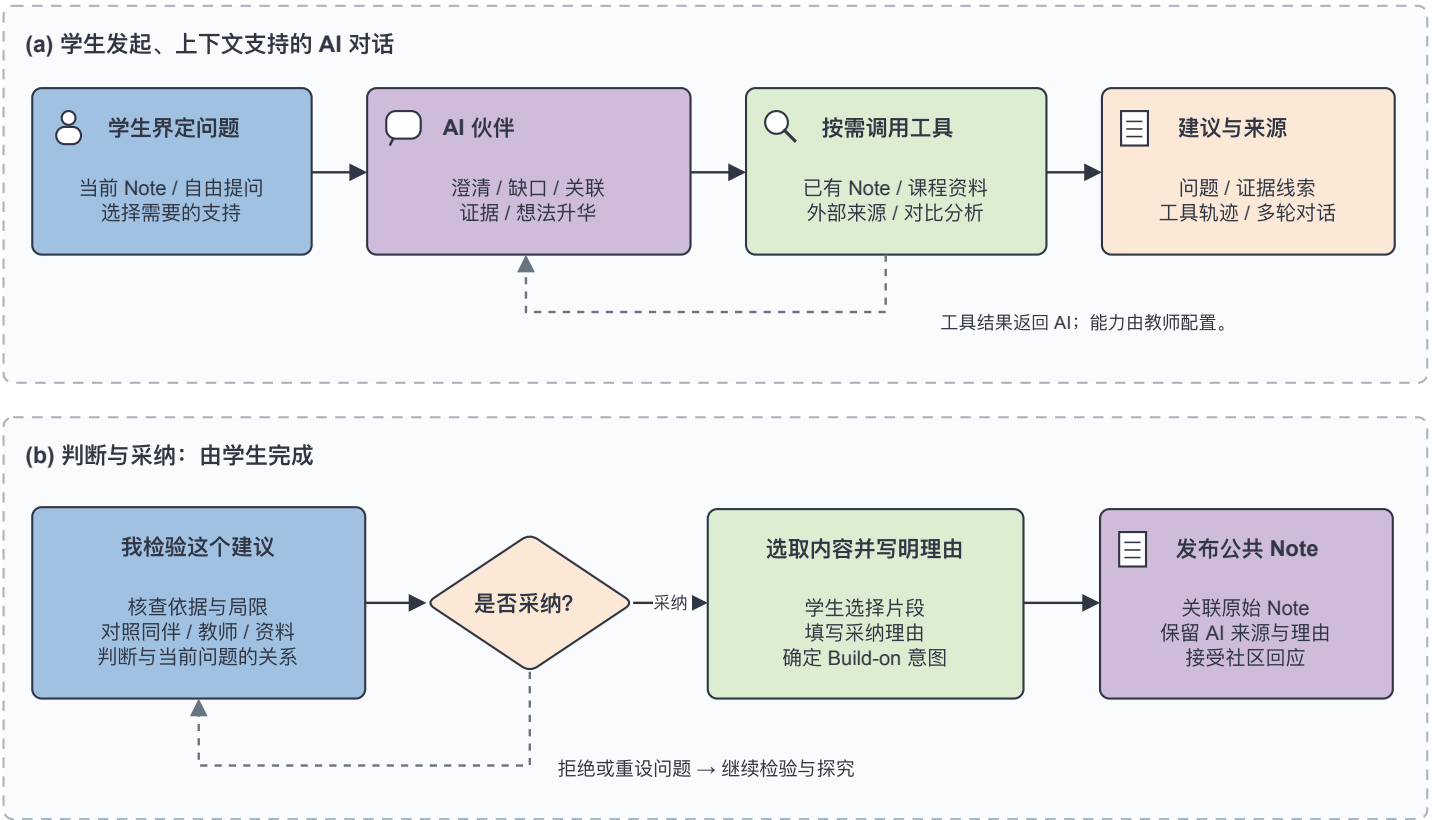
(c) 意图与可追溯性



示意关系方向遵循实现：新 Note → 被回应的既有 Note；颜色仅用于本图。

AI 协作与学生采纳机制

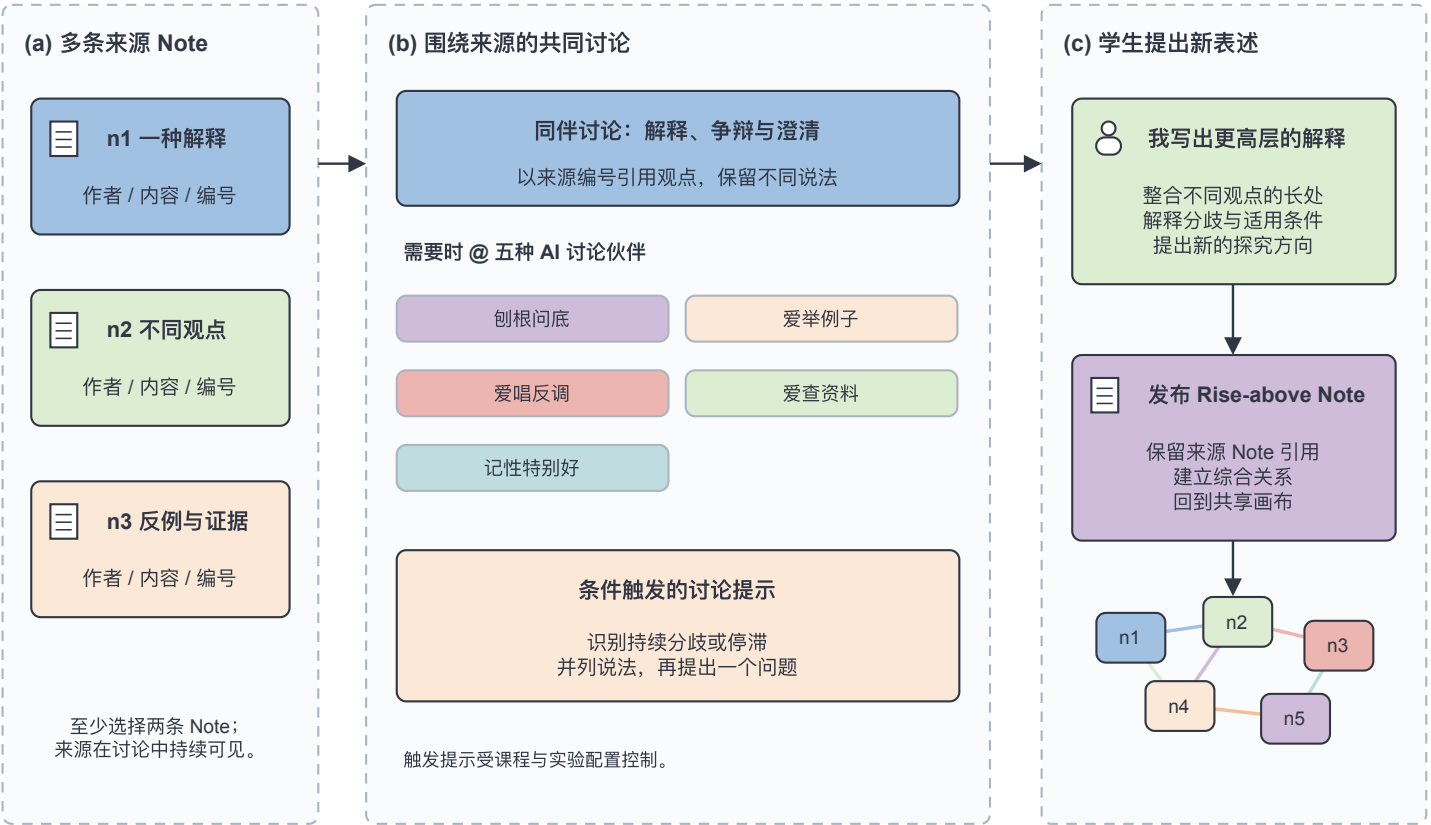
AI 提供可检验的建议，学生保留判断、选择与公共表达的责任



采纳发布接口要求理由；本图描述功能路径，不表示每次 AI 对话都必须被采纳。

从多种观点走向学生自己的综合升华

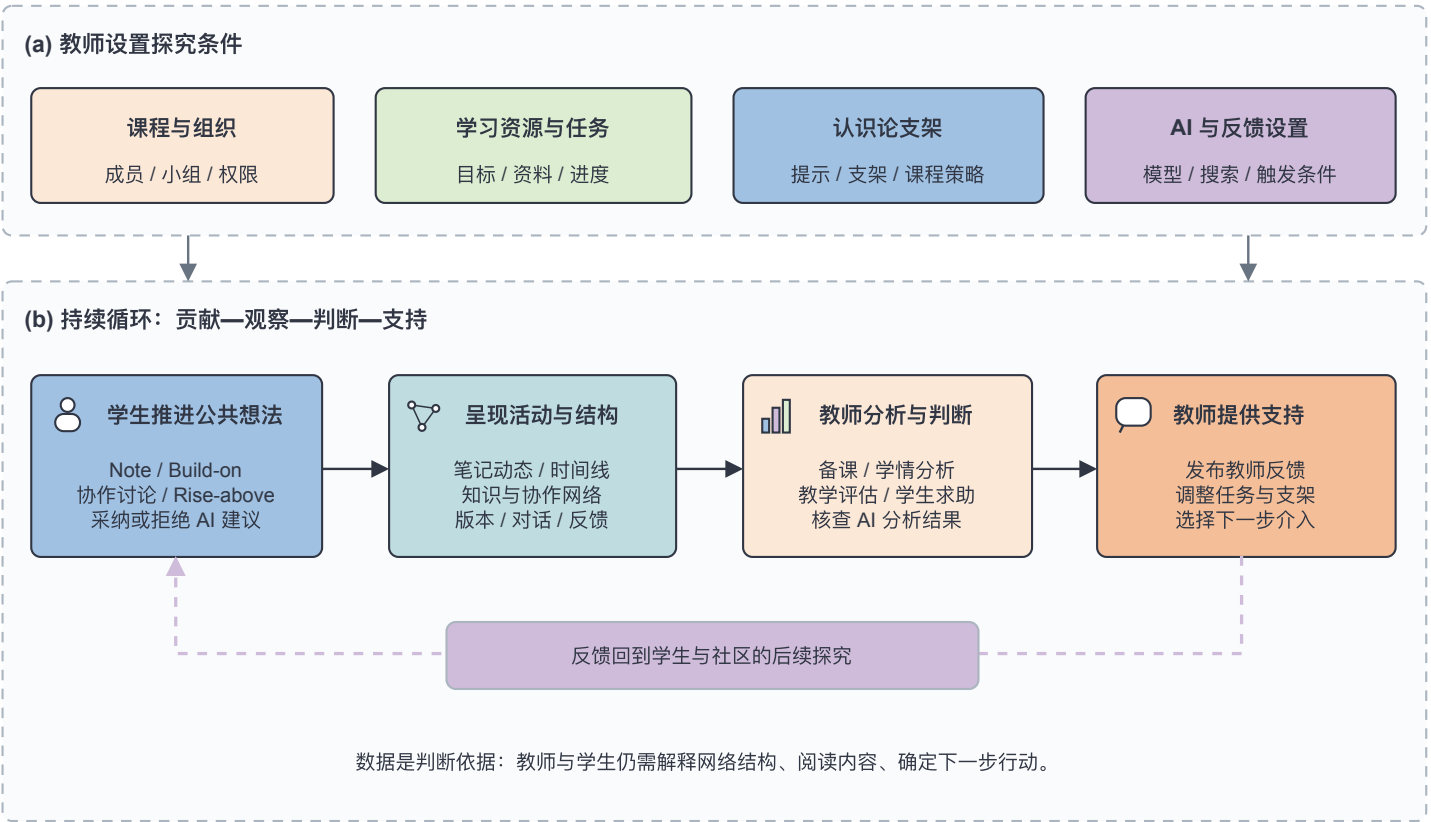
来源常驻、同伴协商、AI 按需参与；高阶表述由学生写出



讨论角色用于提出问题、例子、反例、来源与关联；发布文本来自学生输入。

教师支持与学生探究的反馈循环

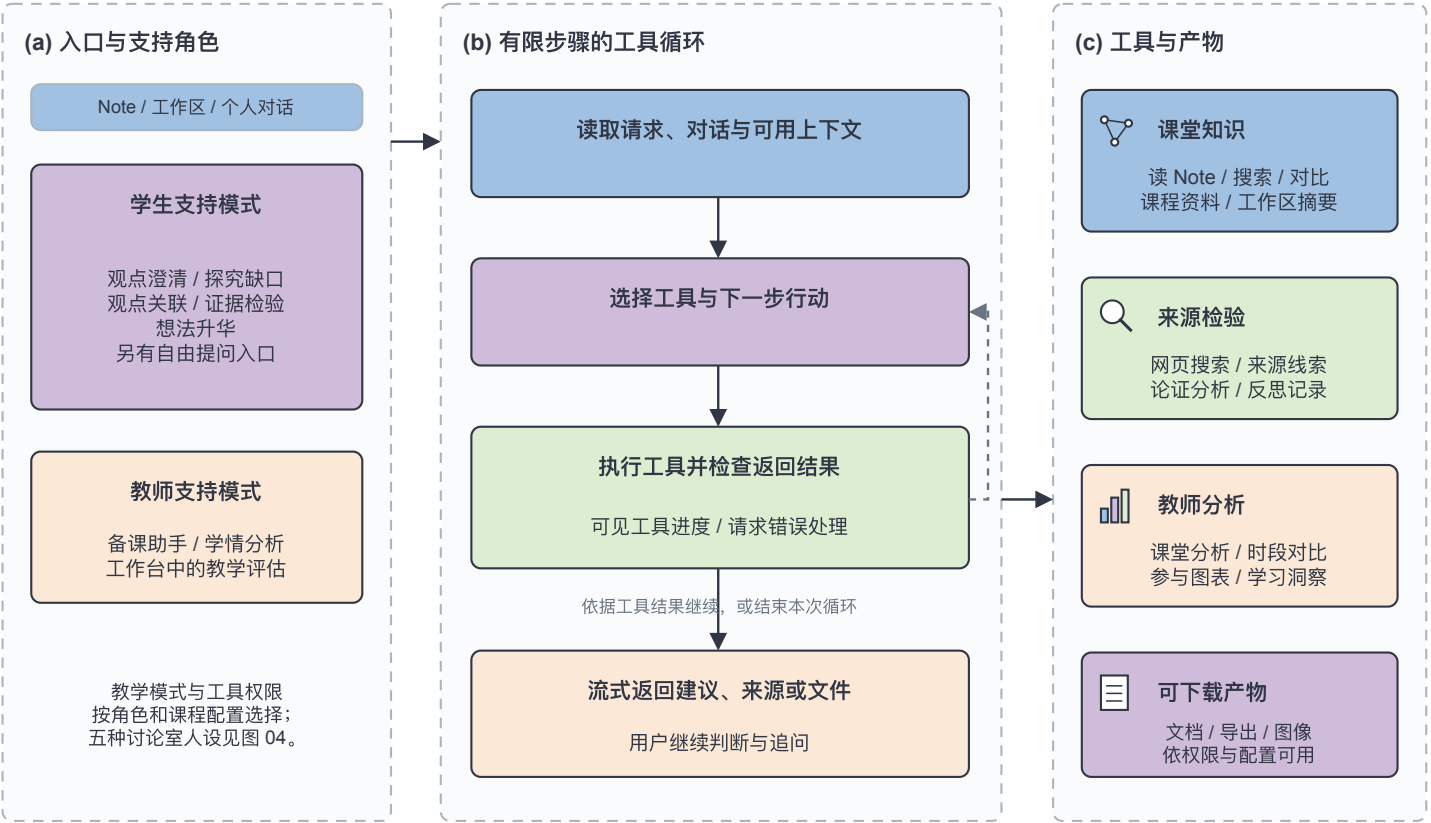
观察社区知识工作，判断介入时机，将反馈带回想法改进



活动、网络、对话和修订数据帮助教师识别需要支持的探究；反馈服务于后续行动。

教学型 AI 角色与工具调用机制

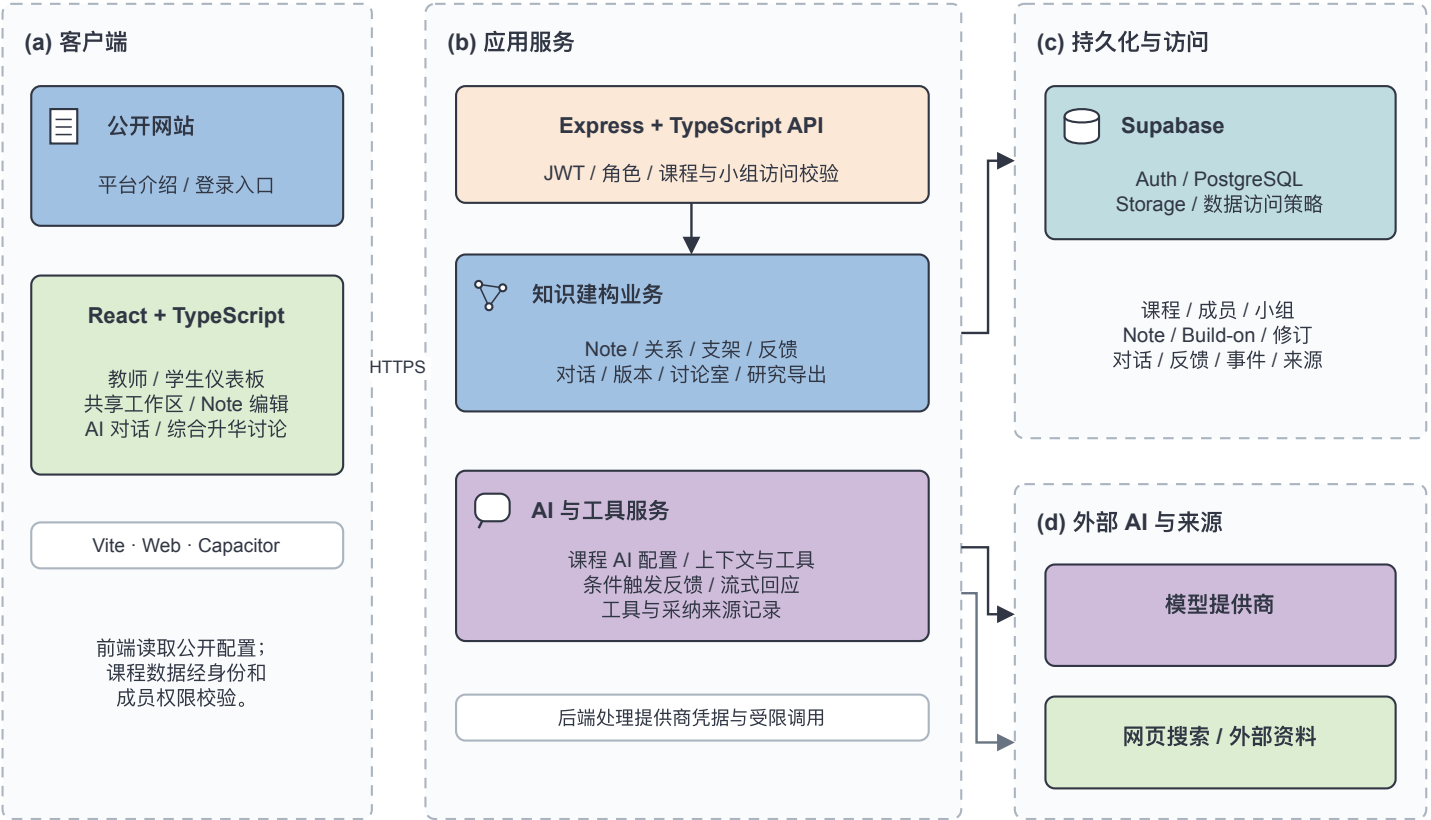
由请求与角色选择工具，在可访问的课程上下文中生成支持



图示为单次请求内的有界工具循环；工具随角色、课程设置和当前入口而变化。

HAKCC 技术架构与数据边界

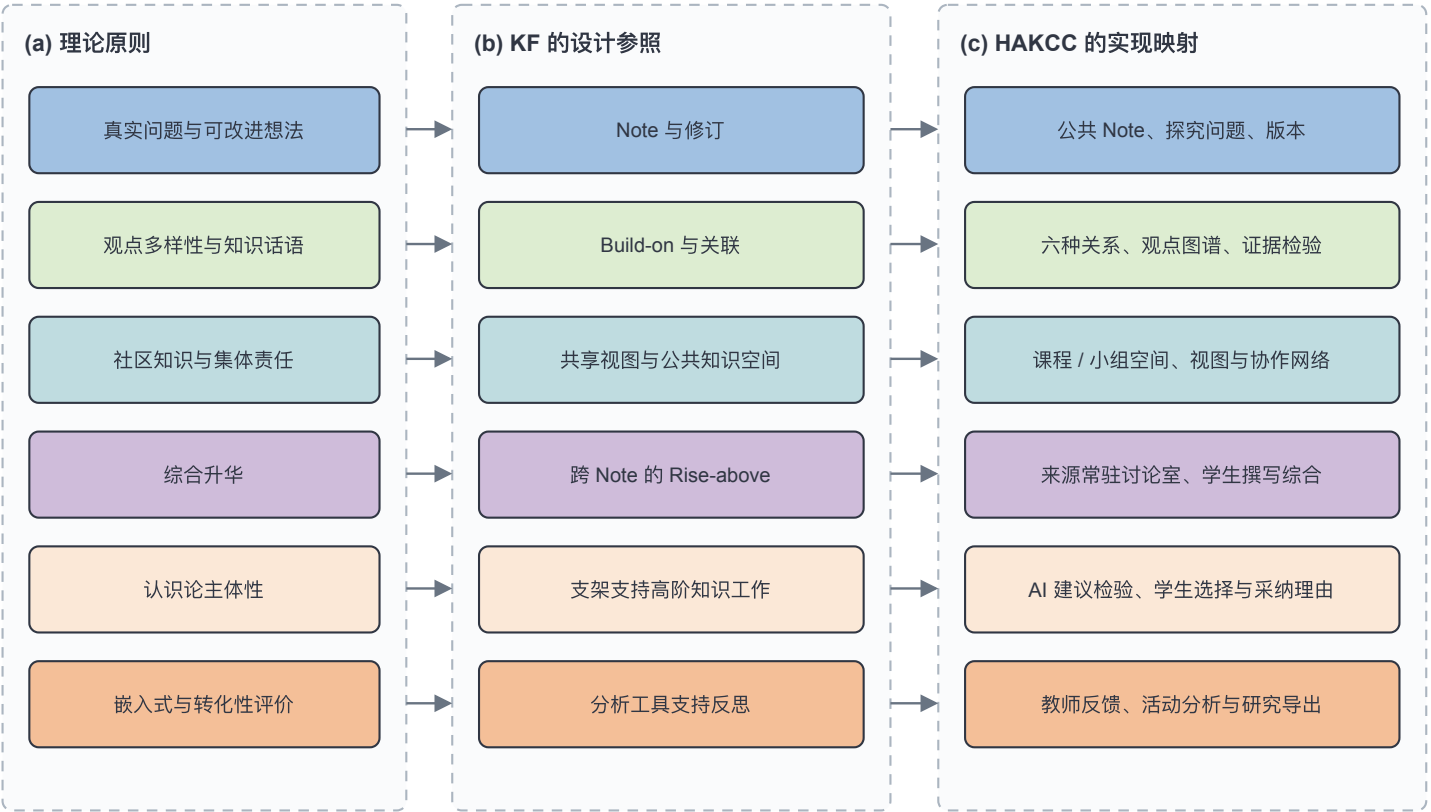
公开界面、课程权限、后端 AI 调用与可追溯知识数据



实线表示请求或数据访问路径；模型与搜索调用由后端代理，课程数据受成员访问校验。

从 Knowledge Building 理论到 HAKCC 设计

以 Scardamalia 与 Bereiter 的理论、Knowledge Forum 的设计为参照

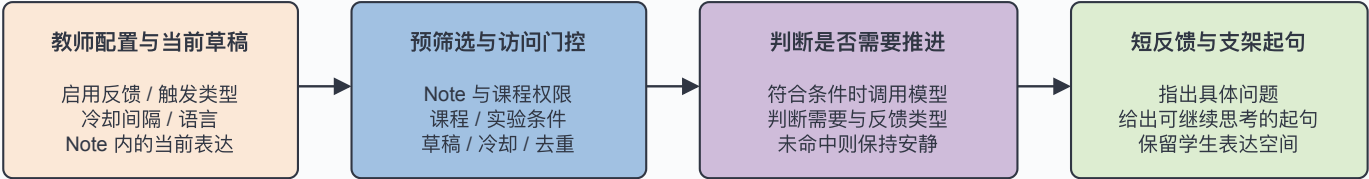


设计映射是解释性对应，非理论成效验证。理论与 KF 参照：Tarchi et al. (2013), 详见图包说明。

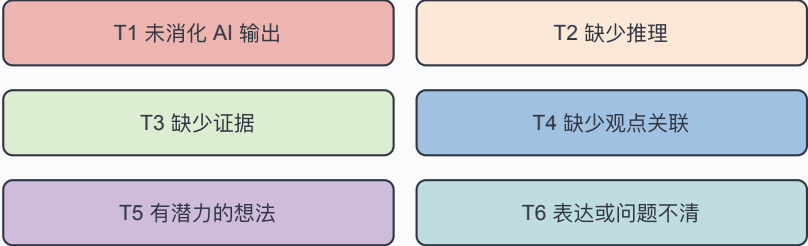
条件触发的 AI 反馈与自适应支架

识别当前表达中需要推进的地方，用短反馈支持学生的下一步思考

(a) 由课程配置与当前 Note 决定反馈路径

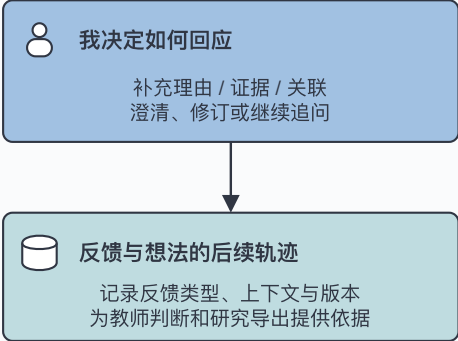


(b) 六类反馈关注点 T1-T6



反馈指出需要推进的地方；T5 关注已有想法的潜力。

(c) 学生响应与可追溯记录



自动检查与主动求助有不同入口；是否呈现反馈依课程设置、实验条件与具体请求而定。